Practical Session 11: RegTech, Compliance & AML

Hands-On: Reciprocal Rank Fusion, the Adverse Media Index, and a Governance Clinic

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Session Overview

Duration: 1 hour (50 min work + 10 min debrief)

Block	Activity	Time
Recap	RRF + Adverse Media Index formulas	8 min
Problem 1	Reciprocal rank fusion by hand	12 min
Problem 2	Compute an Adverse Media Index	12 min
Case study	AMI + RRF in Python (notebook 11)	18 min
Discussion	High-risk classification & HITL design	10 min

Code

All code lives in `code/notebooks/11-regtech-compliance-aml/` (`demo.ipynb`, `gen_rrf_fusion.py`, `exercises.ipynb`). Deterministic — no network, no API key required for the core exercises.

Recap: The Two Formulas You Need

Reciprocal rank fusion (hybrid retrieval):

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{\kappa + \mathrm{rank}_r(d)}, \qquad \kappa = 60.$$

$R = \{\text{dense}, \text{sparse}\}$; missing $\Rightarrow$ rank ∞.

Adverse Media Index (text $\to$ calibrated risk):

$$\mathrm{AMI}(s) = \sigma\left(\alpha + \sum_{i=1}^{n} r_i\, c_i\, t_i\, \mathbf{w}^\top \mathbf{e}_i\right), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

Per document i: relevance r_i, credibility c_i, recency t_i, content score $\mathbf{w}^\top \mathbf{e}_i$; α = baseline intercept.

Recap: Where These Sit in the Pipeline

UNDER THE HOOD

RRF fuses the *retriever*'s two ranked lists; the AMI is computed by the *output formatter* from the reader's per-document evidence scores.

Stage	Produces
Hybrid retriever	dense rank, sparse rank per doc
$\rightarrow$ RRF	fused ranking $\rightarrow$ top-k passages
Reader / ranker	$r_i, c_i, t_i, \mathbf{e}_i$ per doc
$\rightarrow$ AMI	scalar risk $\in [0, 1]$

Why a continuous score? A binary low/high label cannot be compared across subjects, tracked over time, or fed into a quantitative customer-risk model.

Problem 1: Reciprocal Rank Fusion by Hand

Five candidate passages, each with its rank under a dense and a sparse retriever ($\kappa = 60$):

Passage	Dense rank	Sparse rank
court filing	1	2
news article	3	1
press release	2	4
blog post	5	3
registry record	4	5

Task A. Compute $\mathrm{RRF}(d) = \frac{1}{60+\text{dense}} + \frac{1}{60+\text{sparse}}$ for each passage.
Task B. Rank the passages by fused score.
Task C (discussion). Neither "court filing" nor "news article" is ranked #1 by *both* retrievers. Which one wins after fusion, and why does RRF reward cross-signal agreement?

Problem 2: Compute an Adverse Media Index

A screening agent retrieves **three** articles on subject s. Each is scored $(r_i, c_i, t_i, \mathbf{w}^\top \mathbf{e}_i)$:

Article	r_i	c_i	t_i	$\mathbf{w}^\top \mathbf{e}_i$
1 (court doc, recent)	0.9	0.8	1.0	2.0
2 (news, older)	0.6	0.5	0.7	1.2
3 (blog, stale)	0.3	0.9	0.4	0.5

Baseline intercept $\alpha = -2.0$.

Task A. Compute each article's contribution $r_i c_i t_i (\mathbf{w}^\top \mathbf{e}_i)$.
Task B. Sum, add α, apply σ. Report $\mathrm{AMI}(s)$ to 3 d.p.
Task C. Article 3 has the *highest* credibility ($c_3 = 0.9$) yet contributes least. Why? Which factor dominates its low contribution?

Case Study: RRF Fusion in Python (Notebook 11)

From `code/notebooks/11-regtech-compliance-aml/demo.ipynb` — deterministic, no network. **Part 1 — hybrid retrieval fusion:**

```python
import math
KAPPA = 60
docs        = ['court filing','news article','press release'
    ,'blog post','registry record']
dense_rank  = [1, 3, 2, 5, 4]
sparse_rank = [2, 1, 4, 3, 5]

def rrf(ranks):
    return sum(1.0 / (KAPPA + r) for r in ranks)

fused = {d: rrf([dr, sr]) for d, dr, sr in zip(docs,
    dense_rank, sparse_rank)}
for d, s in sorted(fused.items(), key=lambda kv: -kv[1]):
    print(f'{d:16s} RRF={s:.5f}')
```

Case Study: AMI Scoring in Python (Notebook 11)

Part 2 — calibrated Adverse Media Index (same notebook):

```
import math
def sigmoid(x): return 1.0 / (1.0 + math.exp(-x))
alpha = -2.0
# (relevance, credibility, recency, content_score) per
    retrieved article
articles = [(0.9,0.8,1.0,2.0), (0.6,0.5,0.7,1.2),
    (0.3,0.9,0.4,0.5)]
ami = sigmoid(alpha + sum(r*c*t*e for (r,c,t,e) in articles)
    )
print(f'AMI = {ami:.3f}  (probability of genuine AML risk in
     [0,1])')
```

Case Study: Diagnostic Questions

Run the notebook, then answer:

Q1. The fused RRF order puts *court filing* first. Confirm against your Problem 1 hand calculation. Do they match?

Q2. Set `dense_rank` and `sparse_rank` to identical lists (perfect agreement). What happens to the spread of fused scores? What does this say about RRF when both signals agree?

Q3. In the AMI block, raise α from -2.0 to 0.0. How does the score change, and what does α represent operationally (the prior on a random subject being high-risk)?

Q4. Add a *fourth* article with high relevance, high credibility, high recency, and a terrorism-financing content score of 3.0. Recompute the AMI. Why does a single high-severity article move the score so much — and what design risk does that create for calibration of $\mathbf{w}$?

Discussion: Annex III Classification & HITL Design

Part 1 — High-risk or not? For each system, decide the EU AI Act tier and justify:

1. An LLM that *summarises* a news article for a human analyst, no automated decision.
2. An LLM that *auto-declines* onboarding when $\mathrm{AMI} > 0.85$.
3. An LLM that *files* a SAR with FinCEN with no human review.

Part 2 — HITL design. Your bank's reviewers clear 95% of alerts in under 30 seconds (automation bias). Propose *two* concrete changes to the queue or interface that would restore substantive review — referencing task decomposition, uncertainty communication, override feedback, or queue ordering.

Frame your answer around

Degree of automation, individual impact, GDPR Art. 22 review rights, and where a human genuinely outperforms the model.

Solution: Problem 1 — RRF by Hand

Task A. $\mathrm{RRF}(d) = \frac{1}{60+\text{dense}} + \frac{1}{60+\text{sparse}}$:

Passage	Dense	Sparse	RRF
court filing	1	2	$\frac{1}{61} + \frac{1}{62} = 0.03252$
news article	3	1	$\frac{1}{63} + \frac{1}{61} = 0.03227$
press release	2	4	$\frac{1}{62} + \frac{1}{64} = 0.03175$
blog post	5	3	$\frac{1}{65} + \frac{1}{63} = 0.03126$
registry record	4	5	$\frac{1}{64} + \frac{1}{65} = 0.03101$

Task B & C. Order: court filing $\succ$ news article $\succ$ press release $\succ$ blog $\succ$ registry. *News article* is ranked #1 by sparse but only 3rd by dense; *court filing* is #1 dense and #2 sparse — its consistency across *both* signals lifts it above the single-signal winner. That is exactly the robustness property RRF buys.

Solution: Problem 2 — Adverse Media Index

Task A. Contributions $r_i c_i t_i (\mathbf{w}^\top \mathbf{e}_i)$:

$$\begin{aligned}
\text{Art. } 1 &= 0.9 \times 0.8 \times 1.0 \times 2.0 = 1.440 \\
\text{Art. } 2 &= 0.6 \times 0.5 \times 0.7 \times 1.2 = 0.252 \\
\text{Art. } 3 &= 0.3 \times 0.9 \times 0.4 \times 0.5 = 0.054
\end{aligned}$$

Task B. Sum $= 1.746$; add $\alpha = -2.0 \Rightarrow -0.254$:

$$\mathrm{AMI}(s) = \sigma(-0.254) = \frac{1}{1 + e^{0.254}} \approx \mathbf{0.437}.$$

Task C. Despite the highest credibility ($c_3 = 0.9$), Article 3 is crushed by *low relevance* (0.3) and *stale recency* (0.4) — the factors multiply, so any single weak factor drags the whole term down. Credibility cannot rescue a doc that is barely about the subject or years old.

Wrap-Up: Key Takeaways

Problem 1 (RRF):

- Fusion rewards cross-signal agreement — a doc strong on *both* retrievers beats a single-signal #1.
- $\kappa = 60$ compresses rank differences; no learned weighting needed.

Problem 2 (AMI):

- Multiplicative factors mean one weak component (relevance, recency) dominates — credibility alone cannot save a stale, off-target document.
- α sets the prior; **w** encodes offence severity and must be calibrated on enforcement outcomes.

Discussion:

- Degree of automation decides the Annex III tier — *assist* vs. *decide / file*.
- Good HITL gives the human a task they do better than the model.

Next

Lecture 12 — Explainable AI: SHAP, attention attribution, and interpretability methods for the compliance decisions we scored here.

Appendix

Stretch Problems

Extra / optional material — beyond the core lecture.

Appendix: Stretch Problem — Entity Resolution Threshold

A sanctions screener outputs a calibrated match probability $P(\hat{y} = 1 \mid x_1, x_2)$. Onboarding 1,000,000 customers/year; the prevalence of true sanctions hits is ≈ 1 in 50,000.

1. At a threshold giving recall 0.999 and FPR 0.02, how many false positives must analysts review per year? (Hint: $\approx 0.02 \times 10^6$.)
2. Lowering the threshold raises recall toward 1.0 but inflates FPR. Argue, using the cost asymmetry, why sanctions screening still favours the higher recall despite the review burden.
3. How would a two-stage (ANN blocking $\rightarrow$ cross-encoder) pipeline change the operating point versus a single fuzzy-match stage?

Appendix: Stretch Problem — AMI Temporal Decay

Recency weight is $t_i = e^{-\lambda(T - T_i)}$ with $T - T_i$ in days.

1. For $\lambda = 0.01/\text{day}$, compute t_i for an article 30, 180, and 365 days old. Confirm $t \to 0$ slowly.
2. Re-run Problem 2's Article 1, now dated 365 days ago instead of today. Recompute its contribution and the resulting AMI.
3. A subject was *acquitted* 4 years ago but old coverage still surfaces. Argue for an aggressive (large λ) vs. conservative (small λ) decay, and the fairness implications of each.